

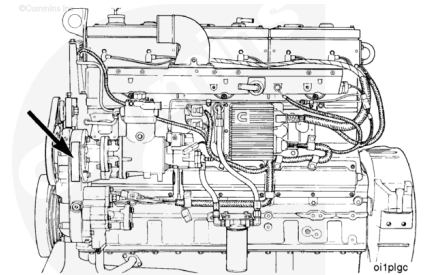
Trainings ▾

Engine Dataplate

The engine dataplate provides the model identification and other important information about the engine.

Have the following engine data available when communicating with a Cummins Authorized Repair Location. The information on the dataplate is **mandatory** when sourcing service parts:

1. Engine serial number (ESN)
2. Control parts list (CPL)
3. Model
4. Advertised horsepower and rpm.



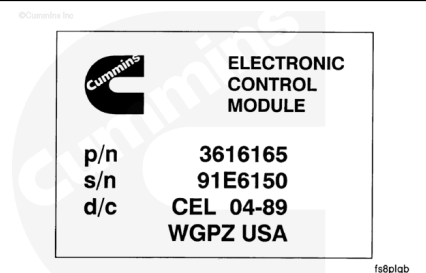
For example, on CELECT™ Plus engines, the electronic control module (ECM) dataplate is located on top of the ECM.

The abbreviations in the accompanying illustration are explained as follows:

p/n = part number

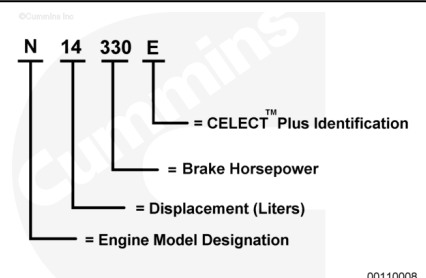
s/n = serial number

d/c = date code.



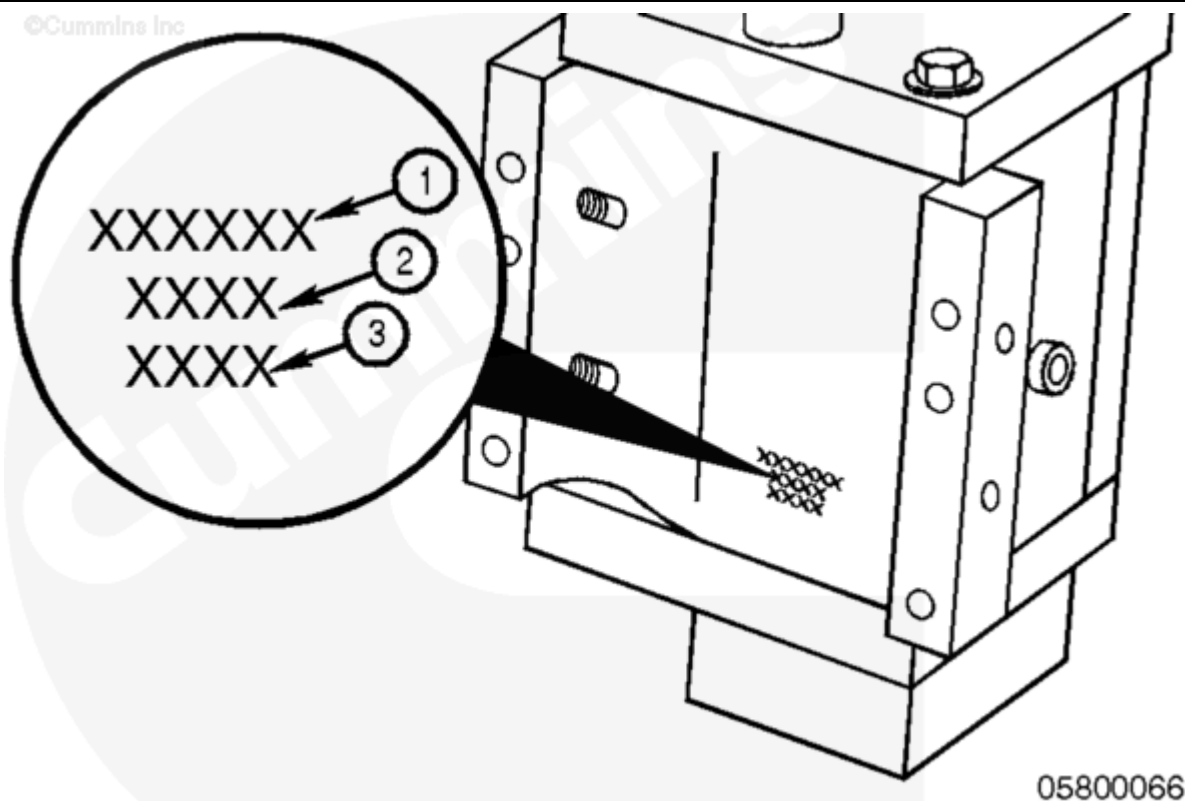
Cummins Engine Nomenclature

The Cummins model name provides information as shown in the accompanying illustration.

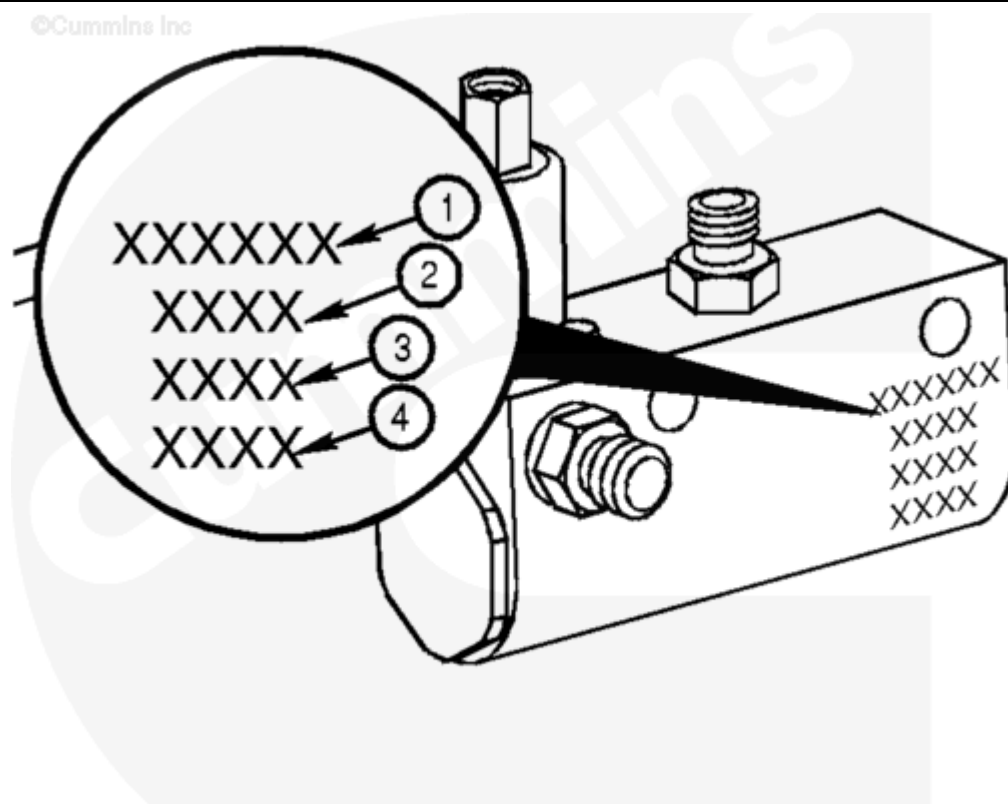


Component Dataplates

Note : There are two dataplates: One is stamped on the Centinel™ oil control valve and the other is stamped onto the Centinel™ control module. Please refer to the model listed on the dataplate when ordering repair/replacement parts.



Dual-Piston



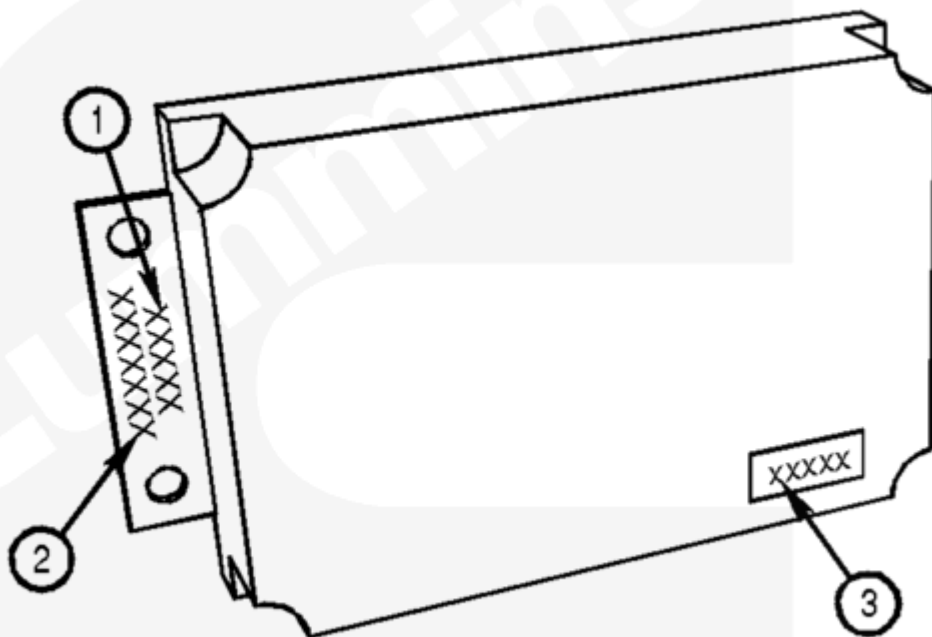
Single-Piston

The part number of the Centinel™ oil control valve is stamped on the valve. The information contained on the oil control valve is:

1. Cummins
2. Part number and voltage
3. Date code.

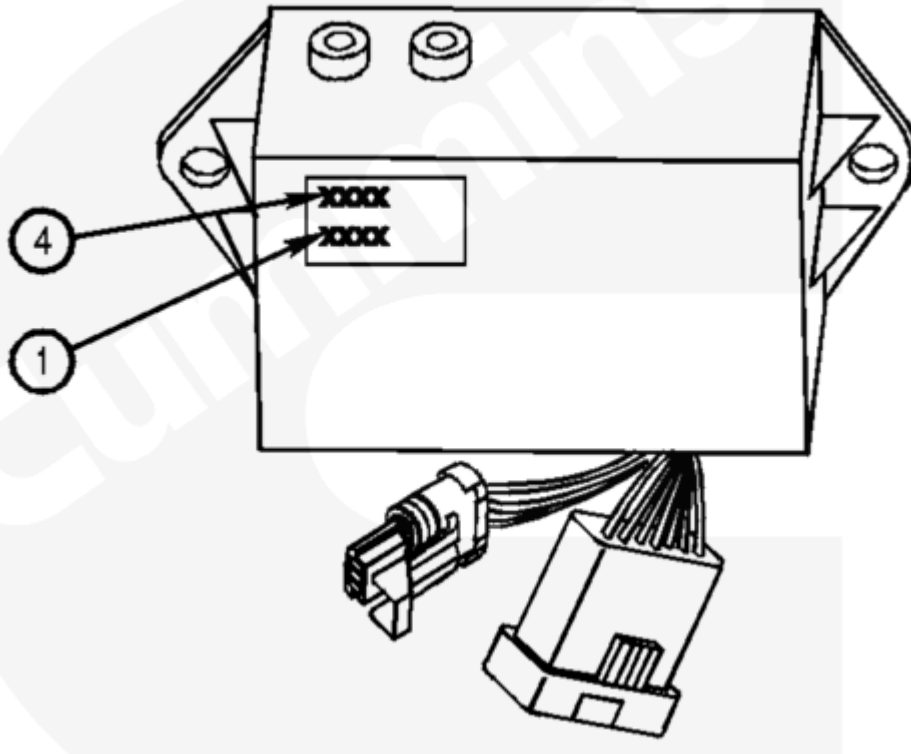
1. Cummins
2. Part number and voltage
3. Supplier identification code
4. Date code.

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High-Horsepower



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Heavy-Duty

The Centinel™ control module has a dataplate located on the front cover. The information contained on the dataplate is:

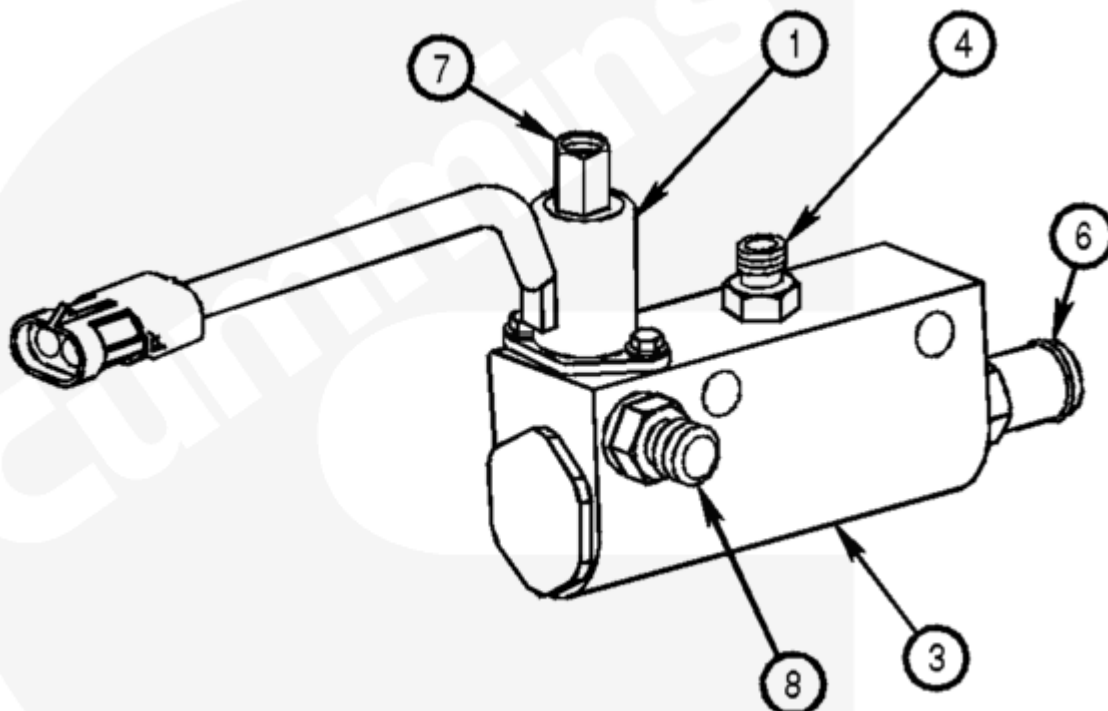
1. Serial number
2. Base module part number.

There are three different Centinel™ control module part numbers available for use on the Centinel™ system. They are:

3. High-Horsepower 12 VDC or 24 VDC, module part number
4. Heavy-Duty 12/24 VDC, module part number.

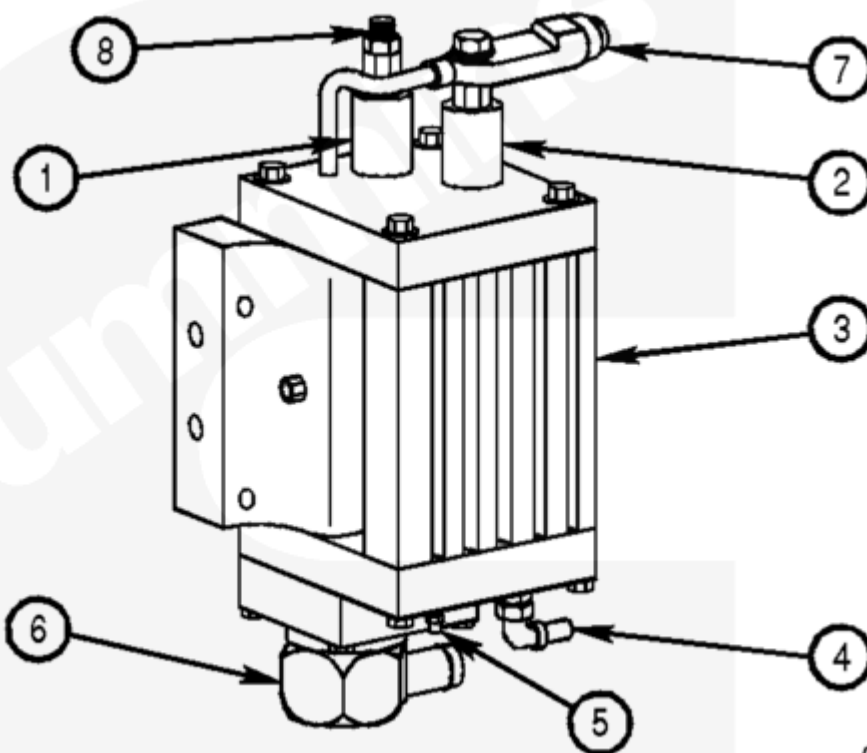
When ordering replacement parts, order by the part number that is specific to your system configuration.

Valve Configurations



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Single-Piston



07800030

Dual-Piston

Centinel™ Oil Control Valve Components

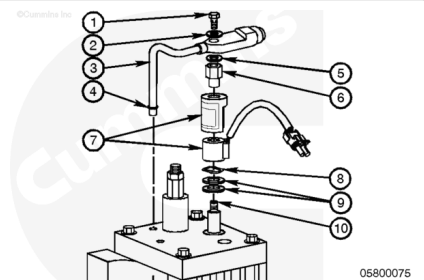
1. Burn solenoid
2. Make-up solenoid
3. Centinel™ oil control valve

4. Oil to engine
5. Air bleed valve
6. Make-up oil supply inlet
7. Engine oil to Centinel™ oil control valve
8. Engine oil to fuel.

Exploded View

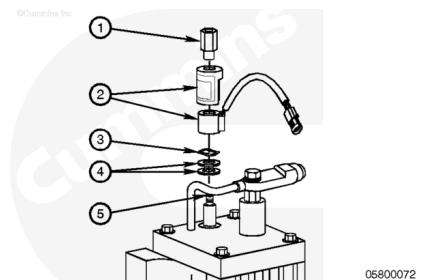
Make-Up Solenoid (dual-piston)

1. Banjo screw
2. Nylon washer
3. Bypass oil tube
4. Bypass oil tube o-ring
5. Nylon washer
6. Solenoid retaining nut
7. Solenoid cover and solenoid coil
8. Spring washer
9. Two washers/spacers
10. O-ring.



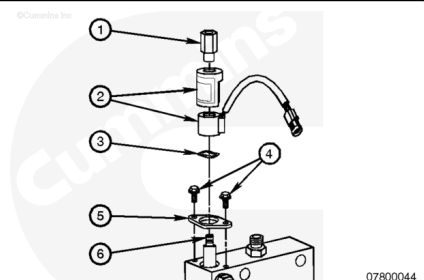
Burn Solenoid (dual-piston)

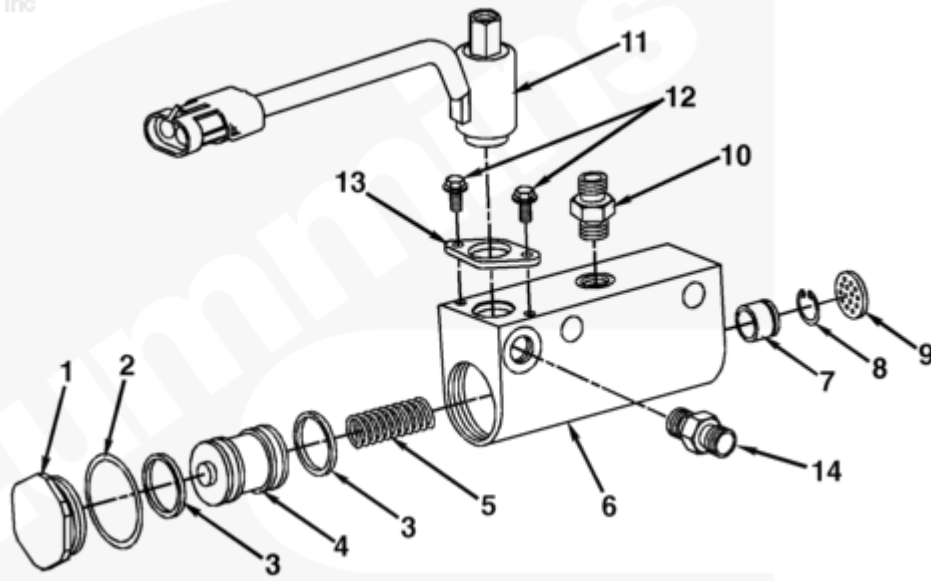
1. Solenoid retaining nut
2. Solenoid cover and solenoid coil
3. Spring washer
4. Two washers/spacers
5. O-ring.



Solenoid (single-piston)

1. Solenoid retaining nut
2. Solenoid cover and solenoid coil
3. Spring washer
4. Screw
5. Clamping plate
6. O-ring.

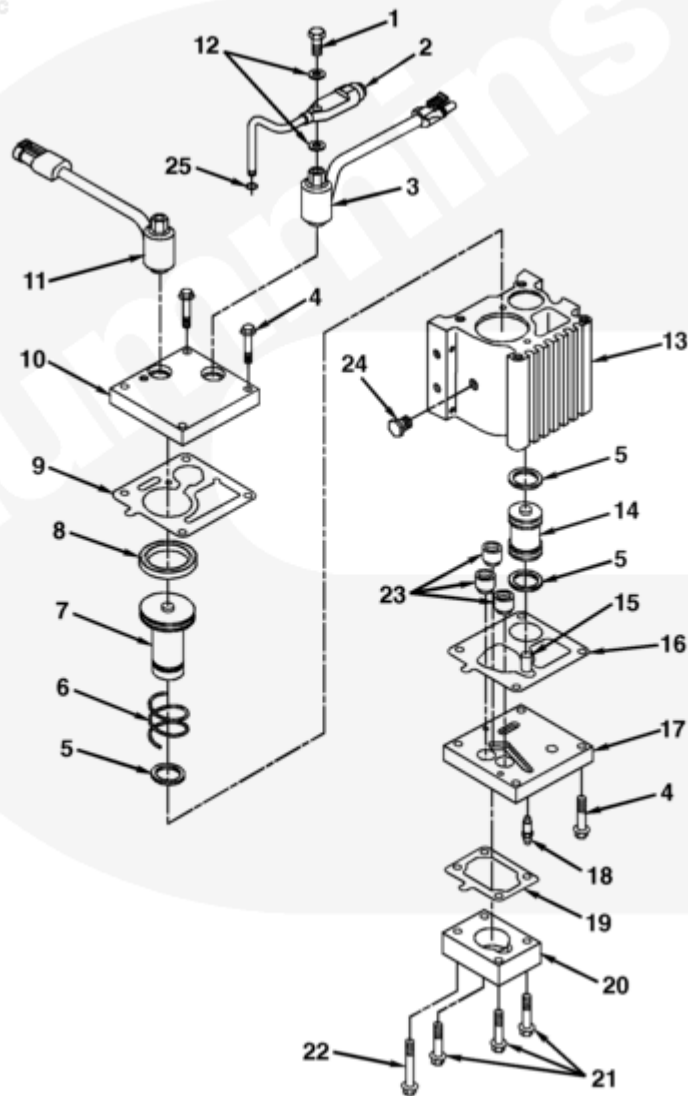




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Oil Control Valve (single-piston)

1. Plug
2. O-ring
3. Seal, u-ring
4. Piston
5. Spring, compression
6. Body, oil control valve
7. Check valve
8. Retaining ring
9. Screen
10. Check valve
11. Valve, solenoid
12. Screw
13. Clamping plate
14. Male union.



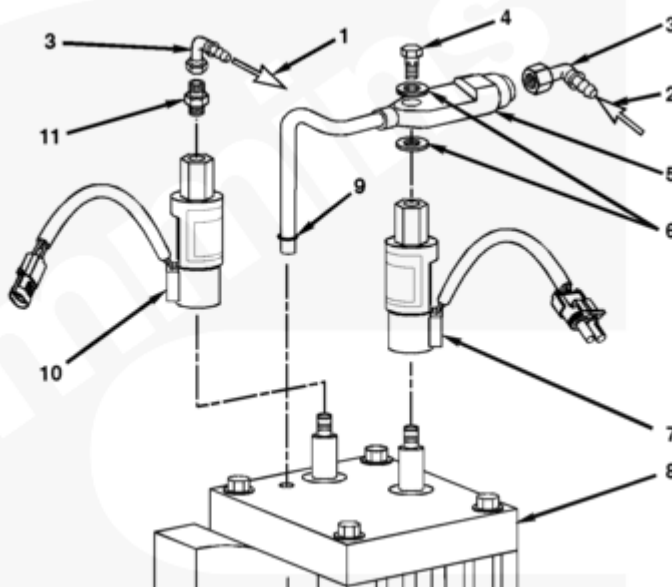
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Oil Control Valve (dual-piston)

1. Screw, banjo
2. Tube, lubricating oil bypass
3. Valve, solenoid (make-up)
4. Capscrew, hex flange head, M6 x 30 (7 places)
5. Seal, u-ring (three places)
6. Spring, compression
7. Piston, oil control valve
8. Seal, u-ring (one place)
9. Gasket, control valve body
10. Cover, oil control valve
11. Valve, solenoid (burn)
12. Washer, banjo screw
13. Body, oil control valve
14. Piston, oil control valve
15. Valve, check
16. Gasket, oil control valve
17. Cover, oil control valve

18. Fitting, bleed screw
19. Gasket, connection
20. Connection, oil supply
21. Capscrew, hex flange head, M6 x 35 (3 places)
22. Capscrew, hex flange head, M6 x 50
23. Valve, check (three places)
24. Fitting, vent
25. O-ring.

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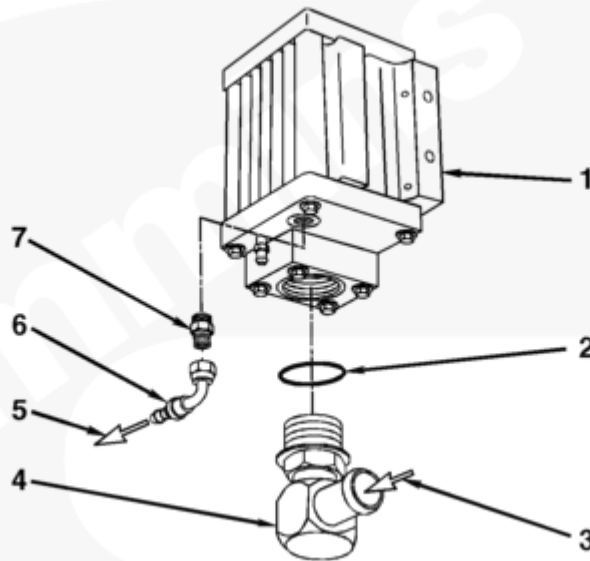


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Centinel™ Dual-Piston Plumbing Assembly (burn-only and burn with make-up - top)

1. Engine oil to fuel return line
2. Engine oil to Centinel™ oil control valve
3. Number 4 face sealing o-ring 90-degree elbow
4. Banjo screw
5. Lubricating oil bypass tube
6. Washer
7. Make-up solenoid
8. Centinel™ oil control valve
9. O-ring
10. Burn solenoid
11. 3/8-24 straight-thread o-ring adapter.

Note : The plumbing on the top of the oil control valve is the same for both the burn-only and burn with make-up systems.

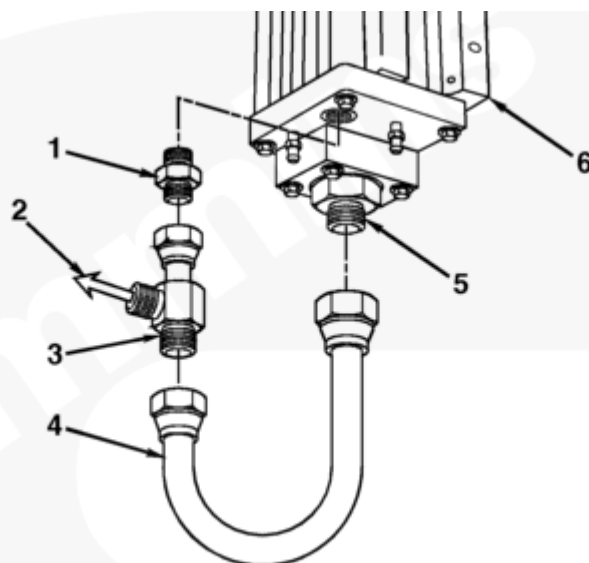


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Centinel™ Dual-Piston Burn with Make-Up Plumbing Assembly (bottom)

1. Centinel™ oil control valve
2. O-ring
3. Make-up oil to Centinel™ oil control valve
4. Make-up oil supply inlet
5. Make-up oil to engine
6. Number 6 face sealing o-ring 90-degree elbow
7. 9/16-18 straight-thread o-ring adapter.

Note : The plumbing on the top of the oil control valve is the same for both the burn-only and burn with make-up systems.



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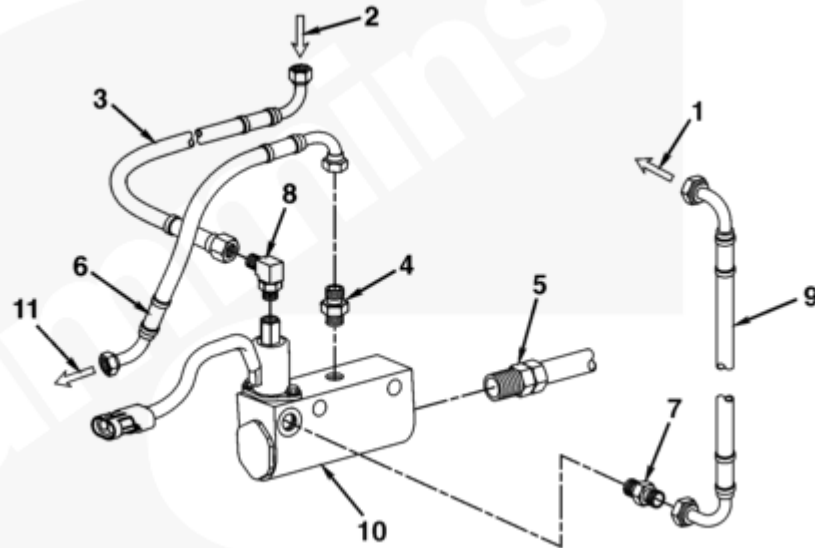
Centinel™ Dual-Piston Burn-Only Plumbing Assembly (bottom)

1. Male union
2. To engine oil sump

3. Male tee with female run
4. Oil transfer tube
5. Male union
6. Dual-piston oil control valve.

Note : The plumbing on the top of the oil control valve is the same for both the burn-only and burn with make-up systems.

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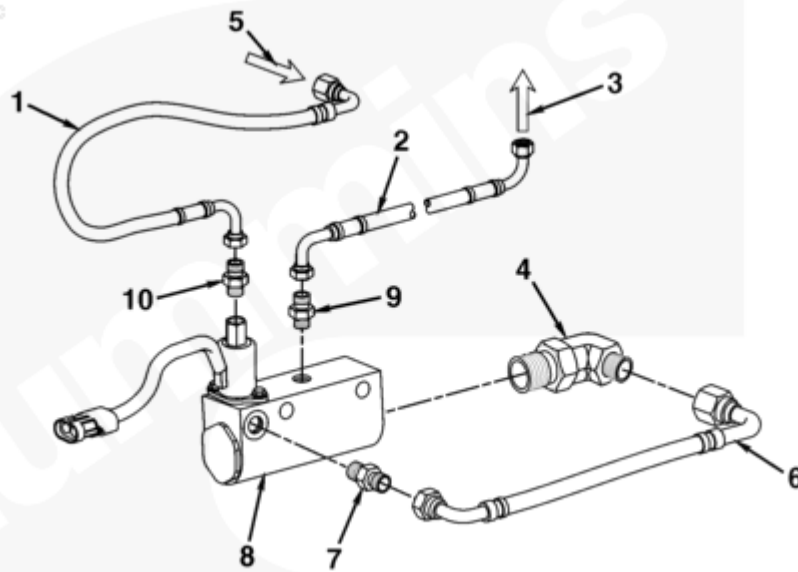


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Single-Piston Burn with Make-Up Plumbing and Mounting Assembly

1. To fuel return
2. Oil supply
3. Oil supply hose
4. Outlet check valve
5. Oil from make-up tank
6. Oil replenishment hose
7. Male union connector
8. Male union elbow
9. Oil burn hose to fuel return
10. Oil control valve
11. Clean oil to engine.

Note : On ISM engines, 3 and 2 are reversed.



07800191

Single-Piston Burn-Only Plumbing and Mounting Assembly (ISM engines)

1. Oil supply hose
2. Oil burn hose to fuel return
3. To fuel return
4. Oil transfer elbow
5. Oil supply
6. Oil transfer hose
7. Male union
8. Oil control valve
9. Outlet check valve
10. Male union.